

Variables and Types of Data:

Variables can be classified as qualitative or quantitative.

Qualitative variables: Qualitative variables are variables that have distinct categories according to some characteristic or attribute.

Example: Gender is a qualitative variable, which can be categorized as male, female, or other.

Quantitative variables: Quantitative variables are variables that can be counted or measured.

Example: the variable age is numerical, and people can be ranked in order according to the value of their ages. Other examples of quantitative variables are heights, weights, and body temperatures.

Quantitative variables can be further **classified into two groups: discrete and continuous.**

Discrete variables: Discrete variables assume values that can be counted.

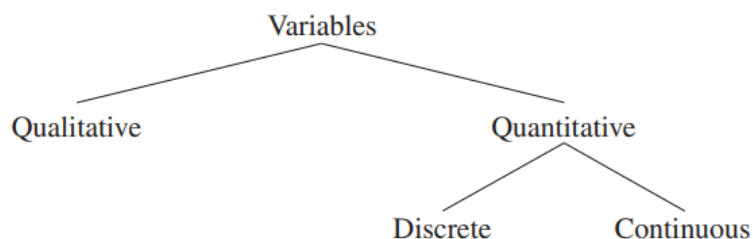
Example: The number of children in a family, the number of students in a classroom, and the number of calls received by a switchboard operator each day for a month.

Continuous variables: Continuous variables can assume an infinite number of values between any two specific values. They are obtained by measuring. They often include fractions and decimals.

Example:

- Height of a person (e.g., 165.4 cm, 170.2 cm)
- Weight of an object (e.g., 55.6 kg, 72.3 kg)
- Temperature (e.g., 36.5°C, 98.7°F)

The classification of variables can be summarized as follows:



EXAMPLE 1–2 Discrete or Continuous Variables

Classify each variable as a discrete variable or a continuous variable.

- a. The highest wind speed of a hurricane
- b. The weight of baggage on an airplane
- c. The number of pages in a statistics book
- d. The amount of money a person spends per year for online purchases

SOLUTION

- a. Continuous, since wind speed must be measured
- b. Continuous, since weight is measured
- c. Discrete, since the number of pages is countable
- d. Discrete, since the smallest value that money can assume is in cents

Class boundaries:

Class boundaries represent the range of values that a rounded data point could fall within before rounding. For example, a recorded height of 73 inches could actually be anywhere between 72.5 and 73.5 inches. Boundaries are written for convenience, like 72.5–73.5, but mean all values up to, but not including, the upper limit.

TABLE 1–1 Recorded Values and Boundaries

Variable	Recorded value	Boundaries
Length	15 centimeters (cm)	14.5–15.5 cm
Temperature	86 degrees Fahrenheit (°F)	85.5–86.5°F
Time	0.43 second (sec)	0.425–0.435 sec
Mass	1.6 grams (g)	1.55–1.65 g

EXAMPLE 1–3 Class Boundaries

Find the boundaries of each variable.

- a. 8.4 quarts
- b. 138 mmHg
- c. 137.63 mg/dL

SOLUTION

- a. 8.35–8.45 quarts
- b. 137.5–138.5 mmHg
- c. 137.625–137.635 mg/dL

- ❖ In addition to being classified as qualitative or quantitative, variables can be classified by how they are categorized, counted, or measured.

Nominal level of measurement: This is the lowest level of measurement, where data are categorized without any order or ranking. The categories are distinct and mutually exclusive.

Example:

- Classifying survey subjects as male or female is another example of nominal-level measurement.
- Classifying residents according to zip codes is also an example of the nominal level of measurement.

Ordinal level of measurement: The ordinal level of measurement classifies data into categories that can be ranked; however, precise differences between the ranks do not exist

Example:

- **Rankings in a competition:** 1st place, 2nd place, 3rd place, etc.
- **Military Rank:** Private, Sergeant, Lieutenant, Captain
- **Educational Level:** High school, Bachelor's, Master's, Doctorate
- Other examples of ordinal data are letter grades (A, B, C, D, F)

Interval level of measurement: The interval level of measurement ranks data, and precise differences between units of measure do exist; however, there is no meaningful zero.

Example:

- **Temperature (Celsius/Fahrenheit):**
Ordered: $10^{\circ}\text{C} < 20^{\circ}\text{C} < 30^{\circ}\text{C}$.
Equal Differences: The difference between 10°C and 20°C is the same as between 20°C and 30°C (both are 10 degrees).
No True Zero: 0°C or 0°F doesn't represent the complete absence of temperature. It's just another point on the scale, not a true zero.
- **IQ Scores:**
Ordered: $80 < 100 < 120$ (ranking based on intelligence level).
Equal Differences: The difference between 100 and 120 is the same as between 80 and 100 (both are 20 points).
No True Zero: An IQ score of 0 doesn't represent "no intelligence," but rather just another score on the scale.

Ratio level of measurement: The ratio level of measurement is the highest level of measurement in statistics. It has all the properties of interval data (ordered, equal differences between values), but with the additional feature of a true zero point.

Example:

- **Height:**

Ordered: 150 cm < 160 cm < 170 cm.

Equal Differences: The difference between 160 cm and 170 cm is the same as the difference between 150 cm and 160 cm (both are 10 cm).

True Zero: A height of 0 cm represents the complete absence of height.

Ratios are Meaningful: Someone who is 180 cm tall is **twice as tall** as someone who is 90 cm tall.

- **Age:**

Ordered: 10 years < 20 years < 30 years.

Equal Differences: The difference between 20 years and 30 years is the same as the difference between 10 years and 20 years (both are 10 years).

True Zero: Age can have a value of 0 years, representing the exact moment of birth.

Ratios are Meaningful: Someone who is 40 years old is **twice as old** as someone who is 20 years old.